final_project

May 6, 2025

[1]:
```
# Step 1

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
from sklearn.preprocessing import StandardScaler
from imblearn.over_sampling import SMOTE
from sklearn.utils import resample
```

[2]:
```
csv_path = "/Users/abirgolam/Downloads/panic_disorder_detection.csv"
df = pd.read_csv(csv_path)

# basic info
print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120000 entries, 0 to 119999
Data columns (total 17 columns):
 #   Column                  Non-Null Count   Dtype
---  ------                  --------------   -----
 0   Participant ID          120000 non-null  int64
 1   Age                     120000 non-null  int64
 2   Gender                  120000 non-null  object
 3   Family History          120000 non-null  object
 4   Personal History        120000 non-null  object
 5   Current Stressors       120000 non-null  object
 6   Symptoms                120000 non-null  object
 7   Severity                120000 non-null  object
 8   Impact on Life          120000 non-null  object
 9   Demographics            120000 non-null  object
 10  Medical History         89776 non-null   object
 11  Psychiatric History     90104 non-null   object
 12  Substance Use           79922 non-null   object
 13  Coping Mechanisms       120000 non-null  object
```

```
 14  Social Support            120000 non-null  object
 15  Lifestyle Factors         120000 non-null  object
 16  Panic Disorder Diagnosis  120000 non-null  int64
dtypes: int64(3), object(14)
memory usage: 15.6+ MB
None
       Participant ID            Age  Panic Disorder Diagnosis
count   120000.000000  120000.000000              120000.00000
mean     50005.474075      43.060983                   0.04285
std      28859.937508      28.226396                   0.20252
min          1.000000     -10.000000                   0.00000
25%      24997.750000      29.000000                   0.00000
50%      49999.500000      42.000000                   0.00000
75%      74980.000000      54.000000                   0.00000
max     100000.000000     500.000000                   1.00000
```

[3]:
```python
print(df.isnull().sum())
numeric_columns = df.select_dtypes(include=['number']).columns
df[numeric_columns] = df[numeric_columns].fillna(df[numeric_columns].mean())

categorical_columns = df.select_dtypes(include=['object']).columns

for column in categorical_columns:
    mode_value = df[column].mode()
    if not mode_value.empty:
        df[column] = df[column].fillna(mode_value.iloc[0])
    else:
        df[column] = df[column].fillna('Unknown')

print(df.isnull().sum())
```

```
Participant ID                0
Age                           0
Gender                        0
Family History                0
Personal History              0
Current Stressors             0
Symptoms                      0
Severity                      0
Impact on Life                0
Demographics                  0
Medical History           30224
Psychiatric History       29896
Substance Use             40078
Coping Mechanisms             0
Social Support                0
Lifestyle Factors             0
Panic Disorder Diagnosis      0
```

```
dtype: int64
Participant ID              0
Age                         0
Gender                      0
Family History              0
Personal History            0
Current Stressors           0
Symptoms                    0
Severity                    0
Impact on Life              0
Demographics                0
Medical History             0
Psychiatric History         0
Substance Use               0
Coping Mechanisms           0
Social Support              0
Lifestyle Factors           0
Panic Disorder Diagnosis    0
dtype: int64
```

[4]:
```
print(df.isnull().sum())
```

```
Participant ID              0
Age                         0
Gender                      0
Family History              0
Personal History            0
Current Stressors           0
Symptoms                    0
Severity                    0
Impact on Life              0
Demographics                0
Medical History             0
Psychiatric History         0
Substance Use               0
Coping Mechanisms           0
Social Support              0
Lifestyle Factors           0
Panic Disorder Diagnosis    0
dtype: int64
```

[5]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# visualize categorical variables

plt.figure(figsize=(8, 5))
sns.countplot(x='Gender', data=df)
```

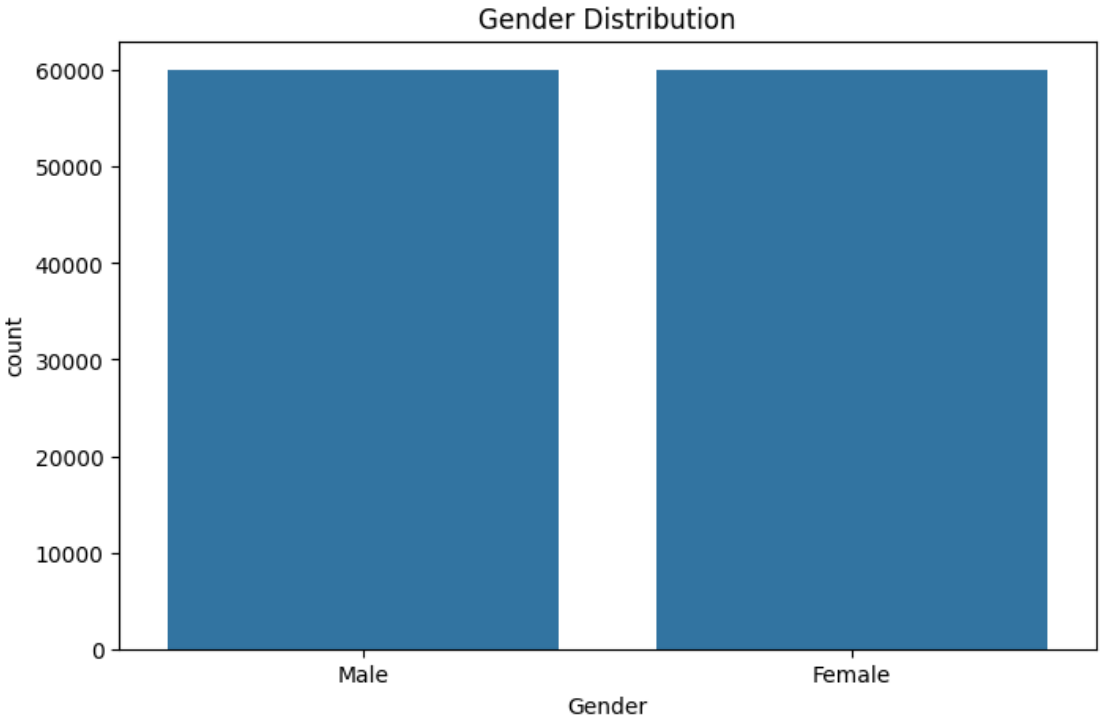
```
plt.title('Gender Distribution')
plt.show()

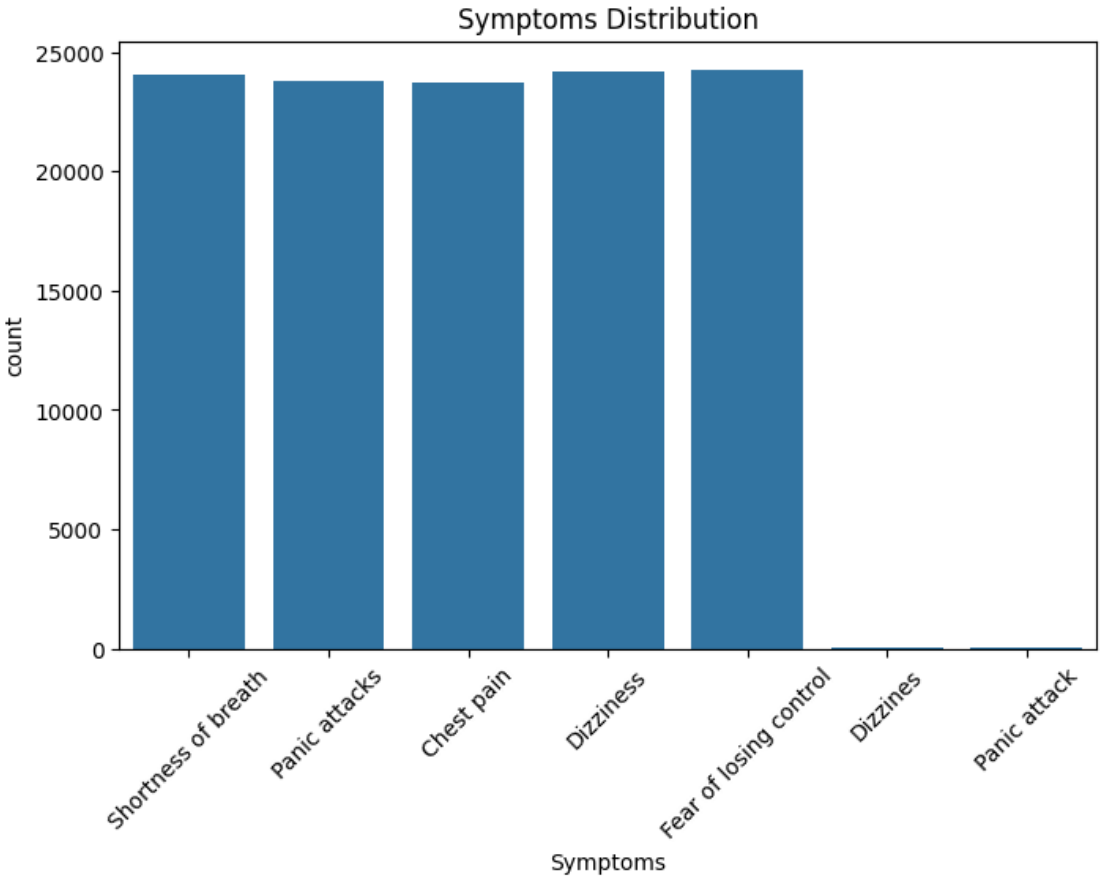

plt.figure(figsize=(8, 5))
sns.countplot(x='Symptoms', data=df)
plt.title('Symptoms Distribution')
plt.xticks(rotation=45)
plt.show()

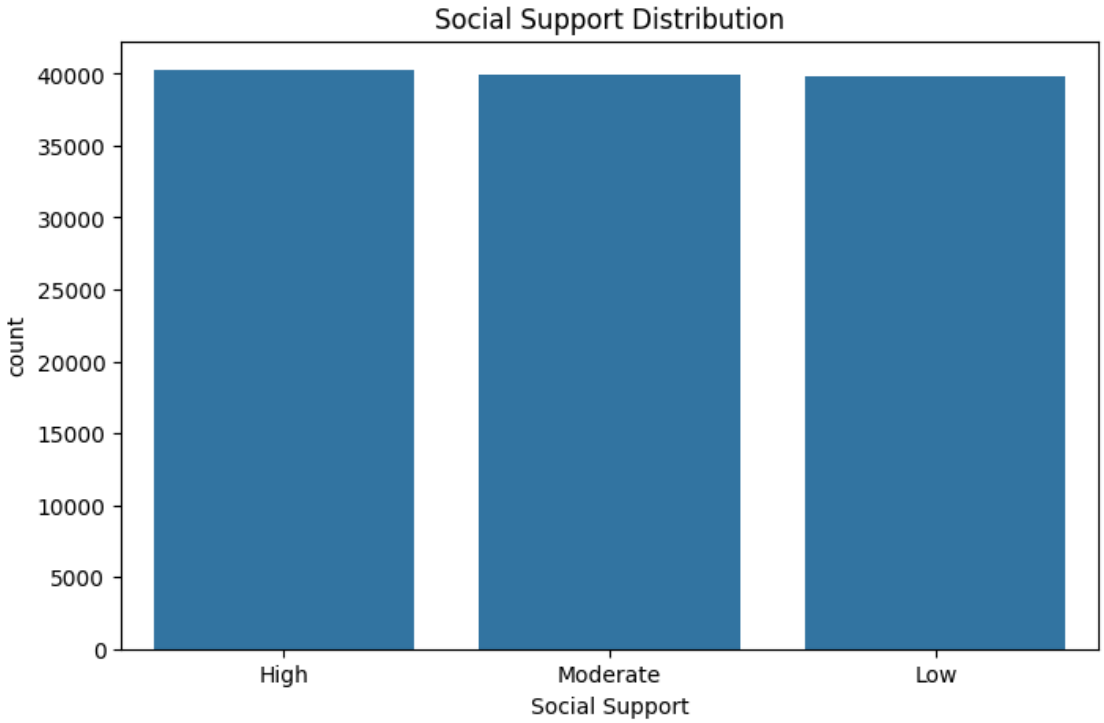

plt.figure(figsize=(8, 5))
sns.countplot(x='Social Support', data=df)
plt.title('Social Support Distribution')
plt.show()

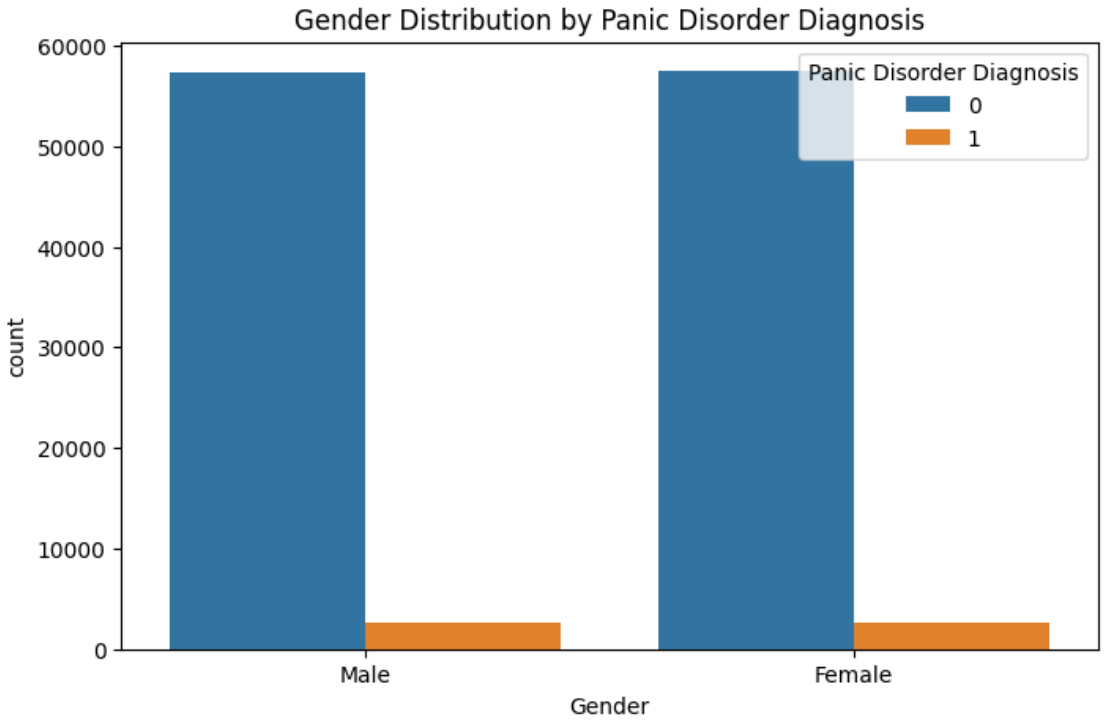

plt.figure(figsize=(8, 5))
sns.countplot(x='Gender', hue='Panic Disorder Diagnosis', data=df)
plt.title('Gender Distribution by Panic Disorder Diagnosis')
plt.show()

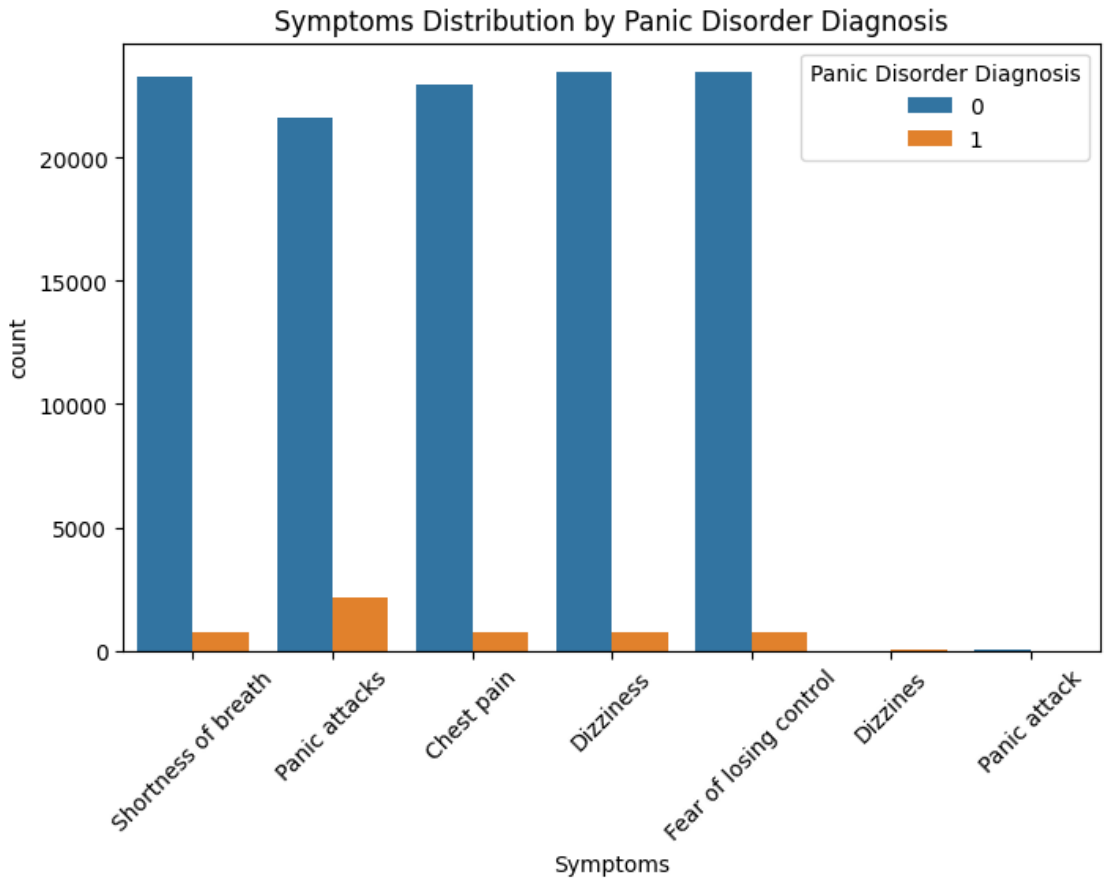
plt.figure(figsize=(8, 5))
sns.countplot(x='Symptoms', hue='Panic Disorder Diagnosis', data=df)
plt.title('Symptoms Distribution by Panic Disorder Diagnosis')
plt.xticks(rotation=45)
plt.show()
```


Gender Distribution
60000
50000
40000
30000
20000
10000
0
count
Male
Female
Gender


Symptoms Distribution
25000
20000
15000
10000
5000
0
count
Shortness of breath
Panic attacks
Chest pain
Dizziness
Fear of losing control
Dizzines
Panic attack
Symptoms


Social Support Distribution
40000
35000
30000
25000
20000
15000
10000
5000
0
count
High
Moderate
Low
Social Support

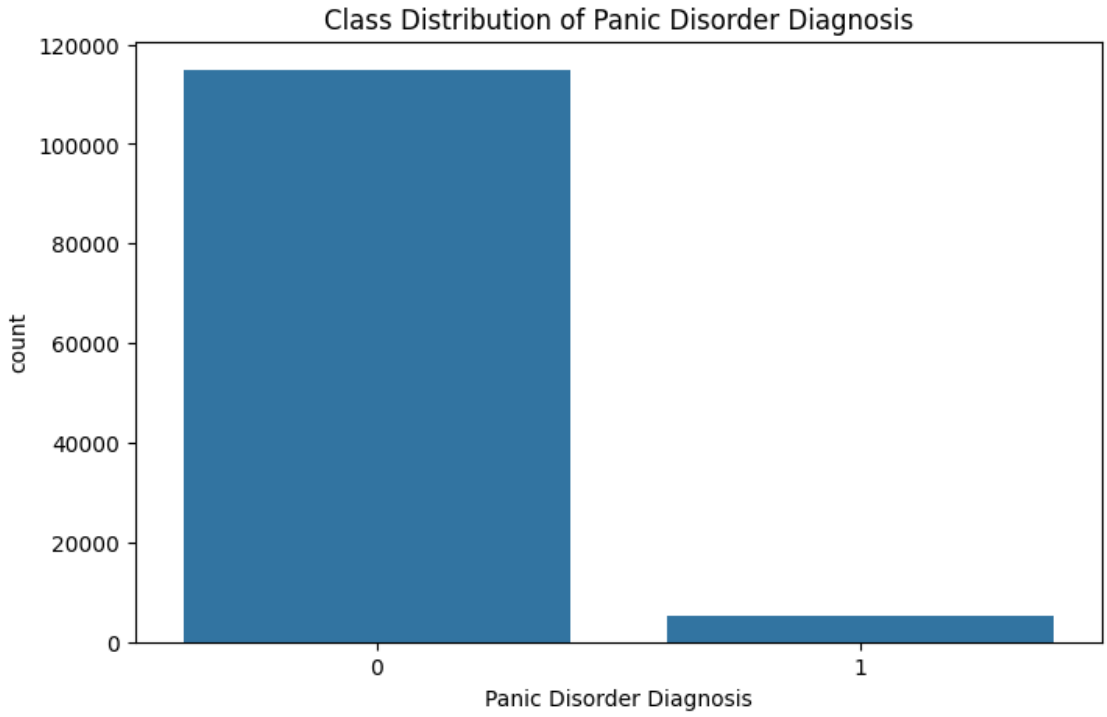

Gender Distribution by Panic Disorder Diagnosis
60000
50000
40000
30000
20000
10000
0
count
Panic Disorder Diagnosis
0
1
Male
Female
Gender



[6]:
```
print(df['Panic Disorder Diagnosis'].value_counts())
```

```
Panic Disorder Diagnosis
0    114858
1      5142
Name: count, dtype: int64
```

[7]:
```
# class imbalance
plt.figure(figsize=(8, 5))
sns.countplot(x='Panic Disorder Diagnosis', data=df)
plt.title('Class Distribution of Panic Disorder Diagnosis')
plt.show()
```



[8]:
```
# step 2

df['Age'] = df['Age'].fillna(df['Age'].median())
df['Medical History'] = df['Medical History'].fillna(df['Medical History'].
 ↪mode()[0])
df['Psychiatric History'] = df['Psychiatric History'].fillna(df['Psychiatric␣
 ↪History'].mode()[0])
df['Substance Use'] = df['Substance Use'].fillna(df['Substance Use'].mode()[0])

print(df.isnull().sum())
```

```
Participant ID          0
Age                     0
Gender                  0
Family History          0
Personal History        0
Current Stressors       0
Symptoms                0
Severity                0
Impact on Life          0
Demographics            0
Medical History         0
Psychiatric History     0
Substance Use           0
```

```
Coping Mechanisms           0
Social Support              0
Lifestyle Factors           0
Panic Disorder Diagnosis    0
dtype: int64
```

[9]:
```
df.drop_duplicates(inplace=True)
print(f"Duplicate entries: {df.duplicated().sum()}")
```

```
Duplicate entries: 0
```

[10]:
```
# categorical columns to lowercase, remove extra spaces
categorical_columns = ['Gender', 'Family History', 'Personal History', 'Current␣
 ↪Stressors',
                       'Symptoms', 'Severity', 'Impact on Life', 'Demographics',
                       'Medical History', 'Psychiatric History', 'Substance␣
 ↪Use',
                       'Coping Mechanisms', 'Social Support', 'Lifestyle␣
 ↪Factors']

for column in categorical_columns:
    df[column] = df[column].str.strip().str.lower()

print(df[categorical_columns].head())
```

```
   Gender Family History Personal History Current Stressors  \
0    male             no              yes          moderate
1    male             no               no              high
2  female            yes               no              high
3  female             no               no          moderate
4    male            yes               no          moderate

              Symptoms  Severity Impact on Life Demographics Medical History  \
0  shortness of breath      mild           mild        rural        diabetes
1        panic attacks      mild           mild        urban          asthma
2        panic attacks      mild    significant        urban        diabetes
3           chest pain  moderate       moderate        rural        diabetes
4        panic attacks      mild       moderate        rural          asthma

   Psychiatric History Substance Use Coping Mechanisms Social Support  \
0     bipolar disorder       alcohol       socializing           high
1     anxiety disorder         drugs          exercise           high
2  depressive disorder       alcohol   seeking therapy       moderate
3     anxiety disorder       alcohol        meditation           high
4     anxiety disorder         drugs   seeking therapy            low

  Lifestyle Factors
0     sleep quality
```

```
1     sleep quality
2          exercise
3          exercise
4     sleep quality
```

[11]:
```
print(df.isnull().sum())
```

```
Participant ID              0
Age                         0
Gender                      0
Family History              0
Personal History            0
Current Stressors           0
Symptoms                    0
Severity                    0
Impact on Life              0
Demographics                0
Medical History             0
Psychiatric History         0
Substance Use               0
Coping Mechanisms           0
Social Support              0
Lifestyle Factors           0
Panic Disorder Diagnosis    0
dtype: int64
```

0.1 Setup: imports & helpers

[26]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, StratifiedKFold,␣
 ↪GridSearchCV
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier

from imblearn.pipeline import Pipeline
from imblearn.over_sampling import SMOTE

from sklearn.metrics import roc_auc_score, classification_report,␣
 ↪ConfusionMatrixDisplay
```

0.2 Step 2 – Data cleaning

[27]:
```
print(df.columns.tolist())
```

```
['Age', 'Gender', 'Family History', 'Personal History', 'Current Stressors',
'Symptoms', 'Severity', 'Impact on Life', 'Demographics', 'Medical History',
'Psychiatric History', 'Substance Use', 'Coping Mechanisms', 'Social Support',
'Lifestyle Factors', 'Panic Disorder Diagnosis']
```

[28]:
```
df.columns = df.columns.str.strip()
```

[29]:
```
# --- Load + minimal cleaning ------------------------------------------
df = pd.read_csv("panic_disorder_detection.csv")
# Clean Age outliers ------------------------------------------------
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")  # ensure numeric
df.loc[(df["Age"] < 0) | (df["Age"] > 100), "Age"] = np.nan

# categorical age bucket
df["Age_Bin"] = pd.cut(
    df["Age"],
    bins=[0, 18, 35, 55, 100],
    labels=["teen", "young", "mid", "senior"]
)
categorical.append("Age_Bin")
df = df.drop(columns="Participant ID")           # unique ID - remove
df.columns = df.columns.str.strip()              # (already done, but safe)

# make sure numeric columns are numeric
numeric = ["Age", "Severity", "Impact on Life"]
df[numeric] = df[numeric].apply(pd.to_numeric, errors="coerce")

# --- Define target and feature groups ---------------------------------
target = "Panic Disorder Diagnosis"

categorical = [c for c in df.columns if c not in numeric + [target]]

# --- Train / test split -----------------------------------------------
X = df.drop(columns=target)
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=42
)

print(f"Train samples: {len(y_train)}   |   Test samples: {len(y_test)}")
print("Positive rate in train:", y_train.mean().round(3),
      " |  Positive rate in test:", y_test.mean().round(3))
```

```
Train samples: 96000   |   Test samples: 24000
```

```
Positive rate in train: 0.043  |  Positive rate in test: 0.043
```

[30]:
```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline   import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from imblearn.over_sampling import SMOTE


numeric_transformer = Pipeline([
    ("imp", SimpleImputer(strategy="median")),
    ("sc",  StandardScaler())
])

categorical_transformer = Pipeline([
    ("imp", SimpleImputer(strategy="most_frequent")),
    ("ohe", OneHotEncoder(handle_unknown="ignore"))
])

preprocess = ColumnTransformer([
    ("num", numeric_transformer, numeric),
    ("cat", categorical_transformer, categorical)
])


def make_pipe(estimator):
    return Pipeline(steps=[
        ("pre", preprocess),
        ("smote", SMOTE(random_state=42)),
        ("clf", estimator)
    ])
```

[31]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="whitegrid")

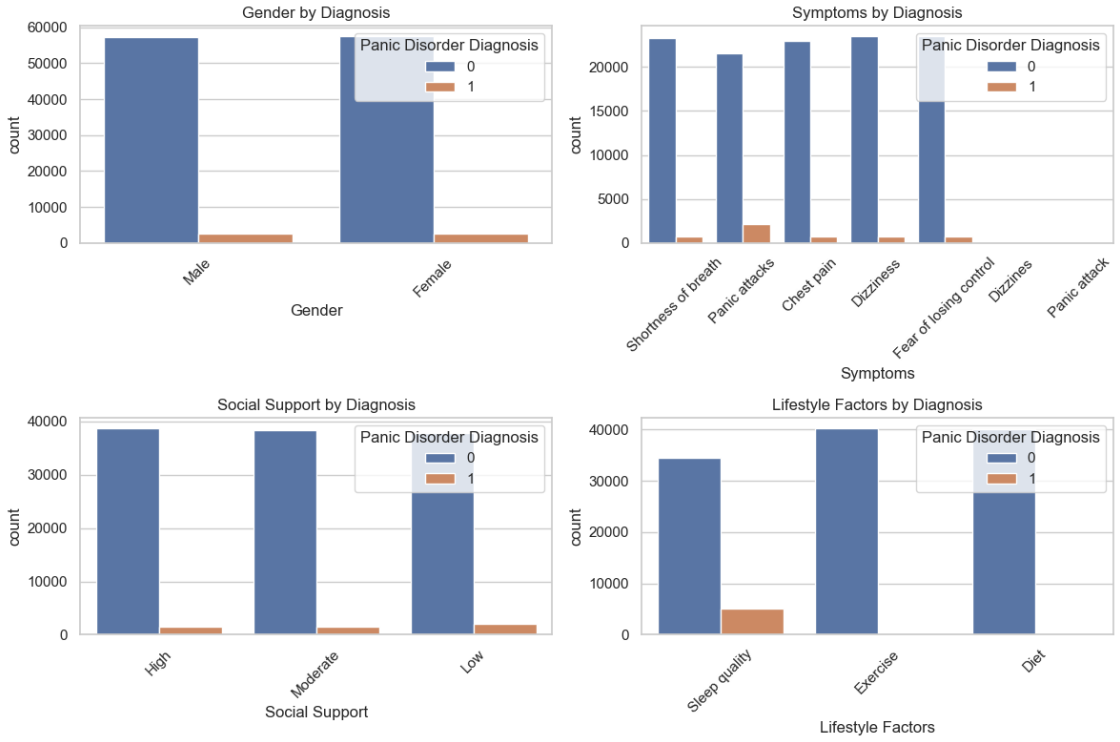

fig, axes = plt.subplots(2, 2, figsize=(12, 8))
cat_vars = ["Gender", "Symptoms", "Social Support", "Lifestyle Factors"]

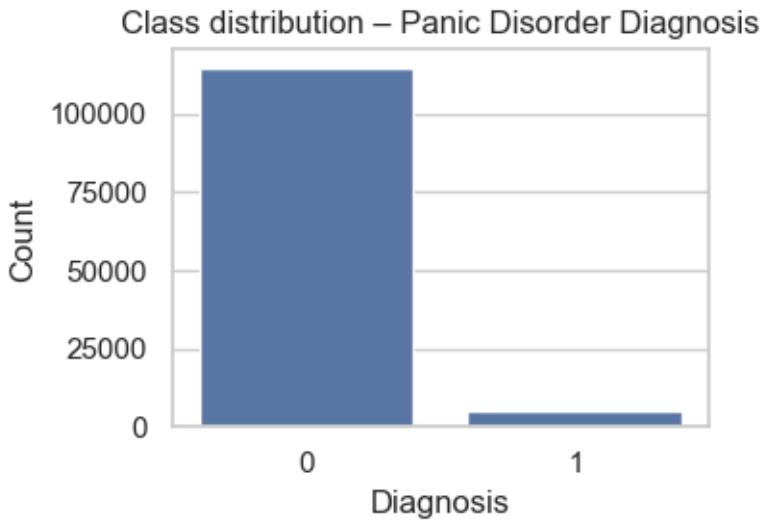
for ax, col in zip(axes.flatten(), cat_vars):
    sns.countplot(data=df, x=col, hue="Panic Disorder Diagnosis", ax=ax)
    ax.set_title(f"{col} by Diagnosis")
    ax.tick_params(axis='x', rotation=45)

plt.tight_layout()
```

```
plt.show()

# imbalance bar-chart
plt.figure(figsize=(4,3))
sns.countplot(x="Panic Disorder Diagnosis", data=df)
plt.title("Class distribution - Panic Disorder Diagnosis")
plt.xlabel("Diagnosis")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```





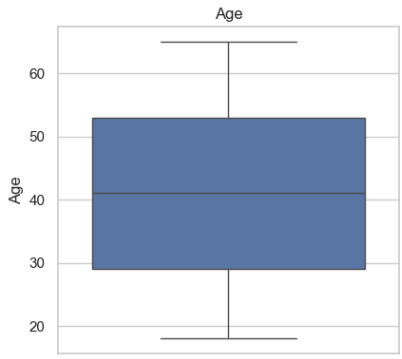
[33]:
```
# 2A. Duplicate check & drop
dupes = df.duplicated().sum()
print(f"Duplicate rows: {dupes}")
if dupes:
    df = df.drop_duplicates().reset_index(drop=True)
    print("Duplicates removed  new shape:", df.shape)
```

```
Duplicate rows: 0
```

[34]:
```
# 2B. Box-plots for numeric outlier
fig, axes = plt.subplots(1, len(numeric), figsize=(4 * len(numeric), 4))
if len(numeric) == 1:
    axes = [axes]

for ax, col in zip(axes, numeric):
    col_data = df[col].dropna()
    if col_data.empty:
        ax.text(0.5, 0.5, f"No data in '{col}'", ha="center", va="center")
        ax.set_axis_off()
    else:
        sns.boxplot(y=col_data, ax=ax)
        ax.set_title(col)

plt.tight_layout()
plt.show()
```



No data in 'Severity'

No data in 'Impact on Life'

[35]:
```
# --- VERIFY AGE CLEANUP & NEW FEATURE --------------------------------

print("Basic stats after cleaning")
print(df["Age"].describe(), "\n")

print("How many Age values are now NaN?", df["Age"].isna().sum())

print("\nAge_Bin value counts (incl. NaNs):")
print(df["Age_Bin"].value_counts(dropna=False), "\n")

print("'Age_Bin' in categorical list →", "Age_Bin" in categorical)

# histogram of cleaned ages
import seaborn as sns
import matplotlib.pyplot as plt
sns.histplot(df["Age"].dropna(), bins=30, kde=False)
plt.title("Age distribution after capping & NaN flagging")
plt.xlabel("Age")
plt.show()
```

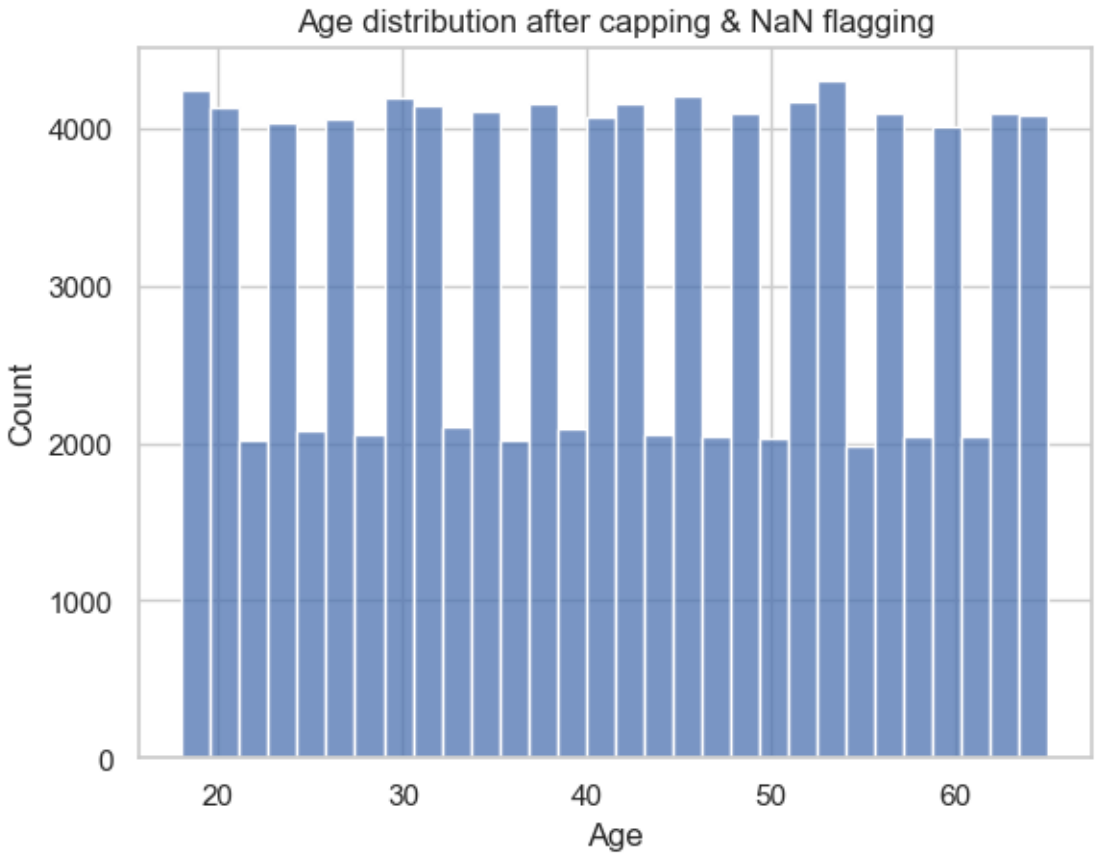
```
Basic stats after cleaning
count    98977.000000
mean        41.453024
std         13.839710
min         18.000000
25%         29.000000
50%         41.000000
75%         53.000000
max         65.000000
Name: Age, dtype: float64

How many Age values are now NaN? 1191

Age_Bin value counts (incl. NaNs):
```

```
Age_Bin
mid       41387
young     35087
senior    20383
teen       2120
NaN        1191
Name: count, dtype: int64

'Age_Bin' in categorical list → True
```



[39]:
```python
# --- Use imblearn's Pipeline that supports over/undersampling ----------
from imblearn.pipeline import Pipeline   # make sure this shadows sklearn's

def make_pipe(estimator):
    return Pipeline(steps=[
        ("pre",   preprocess),                    # ColumnTransformer
        ("smote", SMOTE(random_state=42)),        # oversampler
        ("clf",   estimator)                      # classifier
    ])
```

```
# If you re-executed the notebook and grids vanished, recreate them:
grids = {
    "LR": (
        make_pipe(LogisticRegression(max_iter=500, solver="liblinear")),
        {"clf__C": [0.1, 1, 10]}
    ),
    "RF": (
        make_pipe(RandomForestClassifier(random_state=42)),
        {"clf__n_estimators": [200, 400],
         "clf__max_depth": [None, 15]}
    ),
    "KNN": (
        make_pipe(KNeighborsClassifier()),
        {"clf__n_neighbors": [5, 15],
         "clf__weights": ["uniform", "distance"]}
    )
}
```

[47]:
```
from sklearn.metrics import (accuracy_score, precision_score,
                             recall_score, f1_score, roc_auc_score,
                             RocCurveDisplay)
import pandas as pd
import matplotlib.pyplot as plt

cv   = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
rows = []

for name, (pipe, grid) in grids.items():
    gs = GridSearchCV(pipe, grid, cv=cv, scoring="roc_auc", n_jobs=-1)
    gs.fit(X_train, y_train)

    y_pred = gs.best_estimator_.predict(X_test)
    y_prob = gs.best_estimator_.predict_proba(X_test)[:, 1]

    rows.append({
        "Model": name,
        "Accuracy":  accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred),
        "Recall":    recall_score(y_test, y_pred),
        "F1":        f1_score(y_test, y_pred),
        "AUC":       roc_auc_score(y_test, y_prob)
    })

    print(f"{name}: best CV-AUC = {gs.best_score_:.3f}")

df_results = (pd.DataFrame(rows)
              .set_index("Model")
```

```
            .round(3)
            .sort_values("AUC", ascending=False))
display(df_results)


df_results.to_csv("model_performance_table.csv")


plt.figure(figsize=(6, 5))
for name, (pipe, _) in grids.items():
    fitted = pipe.fit(X_train, y_train)
    RocCurveDisplay.from_estimator(
        fitted, X_test, y_test, name=name, ax=plt.gca())
plt.title("ROC curves (leak-free)")
plt.tight_layout()
plt.show()

# bar chart of F1 & AUC
df_results[["F1", "AUC"]].plot(kind="bar", figsize=(7, 4))
plt.ylabel("Score")
plt.title("F1 vs AUC by Model")
plt.tight_layout()
plt.show()
```

```
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
```

```
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
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LR: best CV-AUC = 0.965

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RF: best CV-AUC = 0.985

/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/joblib/externals/loky/process_executor.py:752: UserWarning: A worker
stopped while some jobs were given to the executor. This can be caused by a too
short worker timeout or by a memory leak.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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```

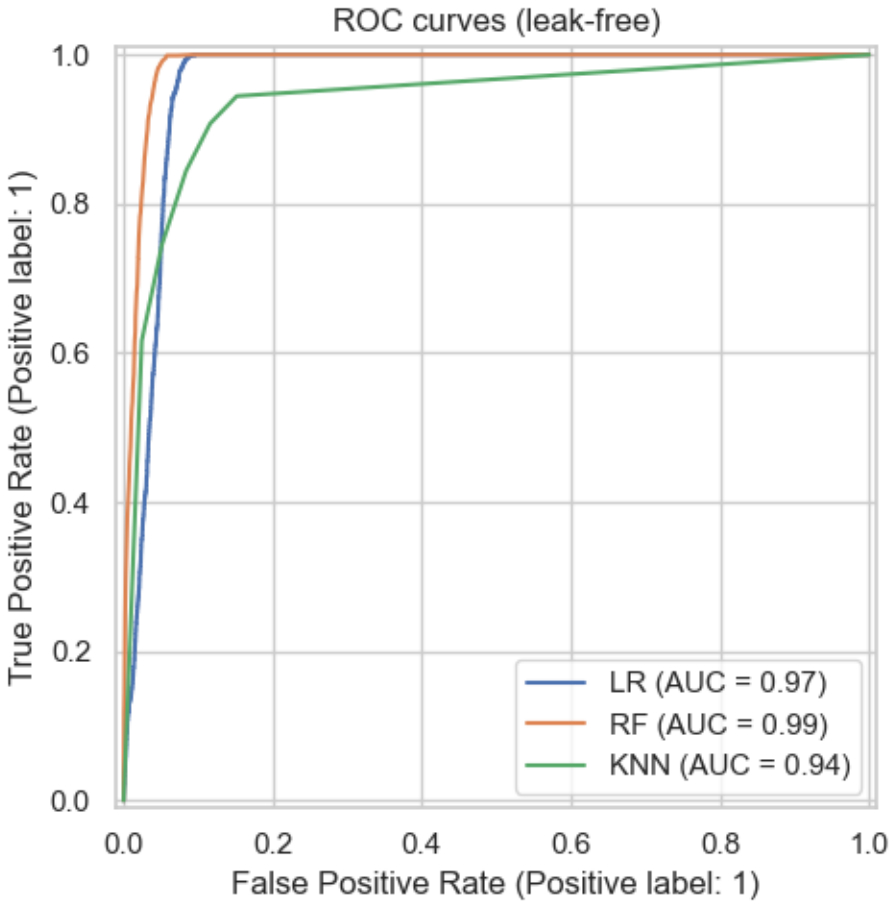
```
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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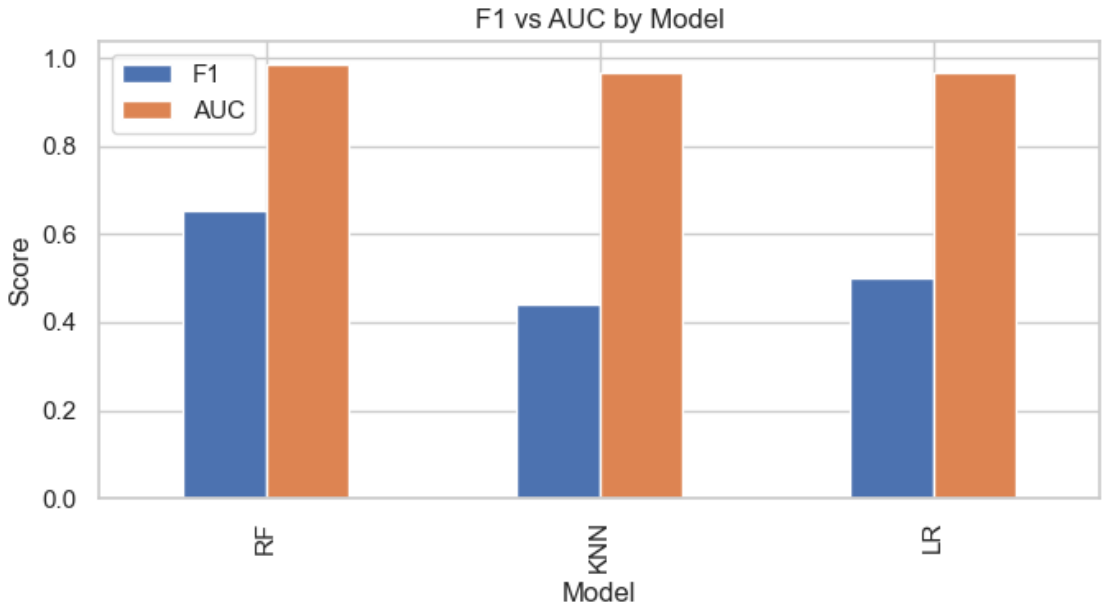
KNN: best CV-AUC = 0.963

       Accuracy  Precision  Recall     F1    AUC
Model
RF        0.970      0.641   0.668  0.655  0.987
KNN       0.897      0.286   0.943  0.439  0.967
LR        0.915      0.336   0.998  0.502  0.965

/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
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/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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is needed for imputation with strategy='median'.
  warnings.warn(
```





[49]:
```
from sklearn.metrics import ConfusionMatrixDisplay, confusion_matrix
import matplotlib.pyplot as plt

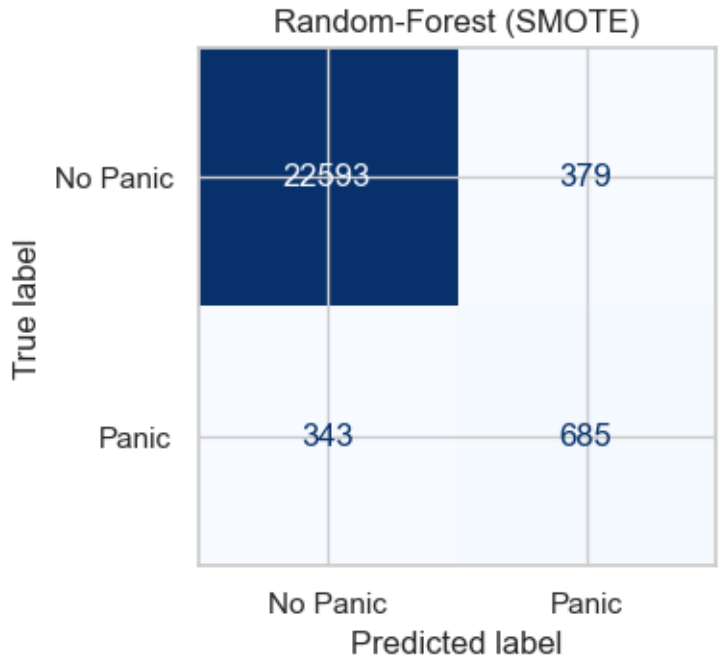
# ------------------------------------------------------------------
# helper to plot a confusion matrix for any fitted pipeline
# ------------------------------------------------------------------
def plot_cm(estimator, X_test, y_test, title):
    y_pred = estimator.predict(X_test)
    cm = confusion_matrix(y_test, y_pred)
    disp = ConfusionMatrixDisplay(cm, display_labels=["No Panic", "Panic"])
    fig, ax = plt.subplots(figsize=(4, 4))
    disp.plot(ax=ax, cmap="Blues", colorbar=False, values_format="d")
    ax.set_title(title)
    plt.tight_layout()
    plt.show()
    print(f"{title}     TN:{cm[0,0]}  FP:{cm[0,1]}  FN:{cm[1,0]}  TP:
 ↪{cm[1,1]}\n")

# ------------------------------------------------------------------
# use your *best* fitted estimators from the grid-search loop
# ------------------------------------------------------------------
best_lr = grids["LR"][0].fit(X_train, y_train)
best_rf = grids["RF"][0].fit(X_train, y_train)
best_kn = grids["KNN"][0].fit(X_train, y_train)

plot_cm(best_rf, X_test, y_test, "Random-Forest (SMOTE)")
```

```
plot_cm(best_lr, X_test, y_test, "Logistic Regression (SMOTE)")
plot_cm(best_kn, X_test, y_test, "K-Nearest Neighbours (SMOTE)")
```

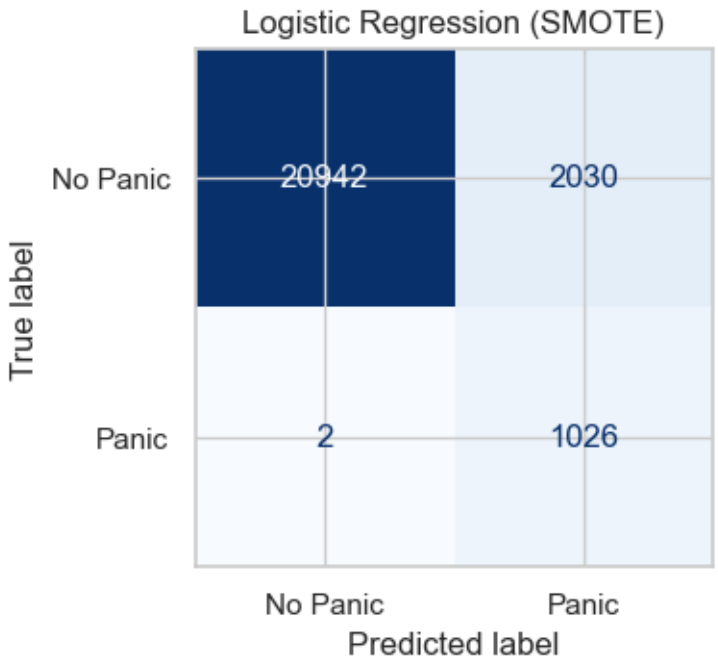
```
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  warnings.warn(
```




```
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```

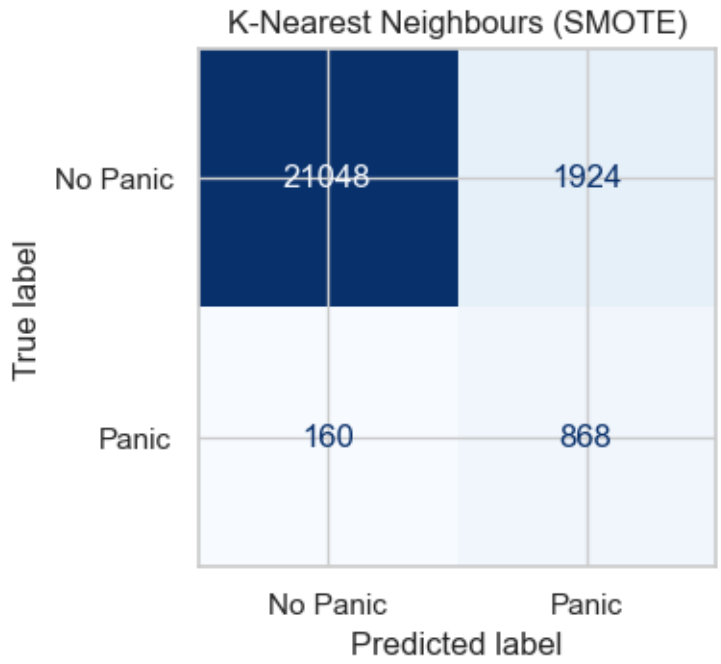
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observed values: ['Severity' 'Impact on Life']. At least one non-missing value
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/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/utils/extmath.py:205: RuntimeWarning: invalid value encountered
in matmul
  ret = a @ b
```

```
Random-Forest (SMOTE)     TN:22593  FP:379  FN:343  TP:685
```




```
Logistic Regression (SMOTE)     TN:20942  FP:2030  FN:2  TP:1026
```

```
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
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  warnings.warn(
```




```
K-Nearest Neighbours (SMOTE)     TN:21048  FP:1924  FN:160  TP:868
```

[48]:
```
ohe_cols = (preprocess.named_transformers_["cat"]
            .named_steps["ohe"]
            .get_feature_names_out(categorical))
feature_names = np.concatenate([numeric, ohe_cols])

# 1  Random-Forest
rf_best = make_pipe(RandomForestClassifier(random_state=42,
                                           n_estimators=400,
                                           max_depth=None))
rf_best.fit(X_train, y_train)
imp = rf_best.named_steps["clf"].feature_importances_
top_idx = np.argsort(imp)[-15:]

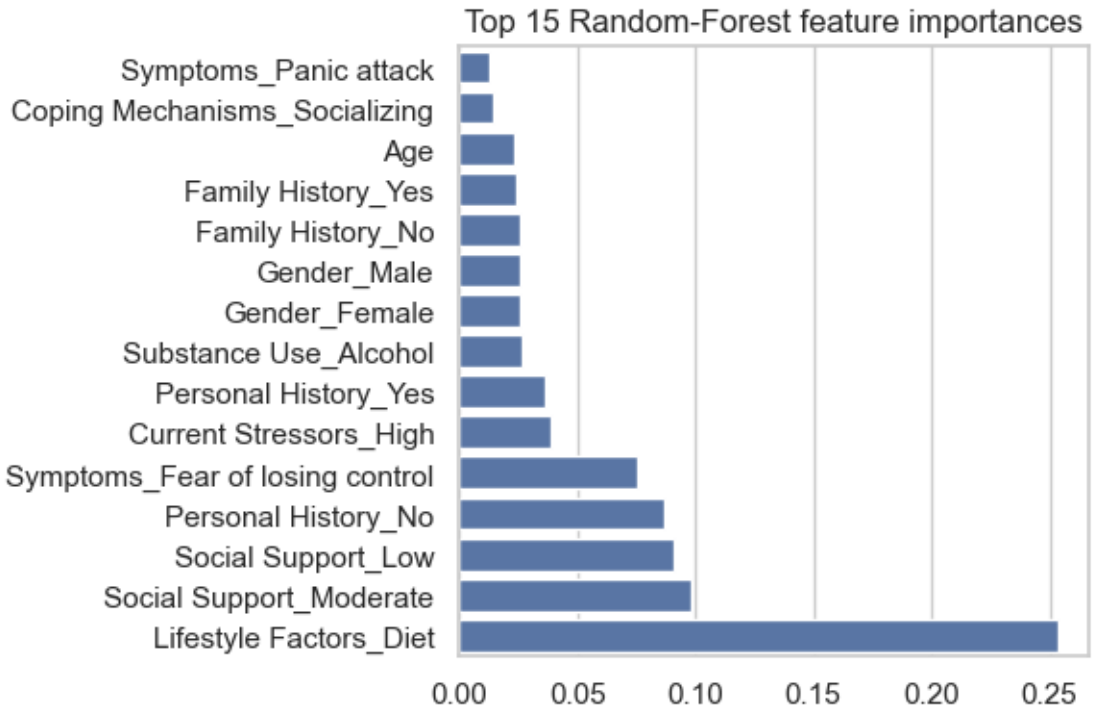
plt.figure(figsize=(6, 4))
sns.barplot(x=imp[top_idx], y=feature_names[top_idx])
plt.title("Top 15 Random-Forest feature importances")
plt.tight_layout()
plt.show()

# 2  logistic reg coefficents
log_best = make_pipe(LogisticRegression(max_iter=500, solver="liblinear", C=1))
log_best.fit(X_train, y_train)
```

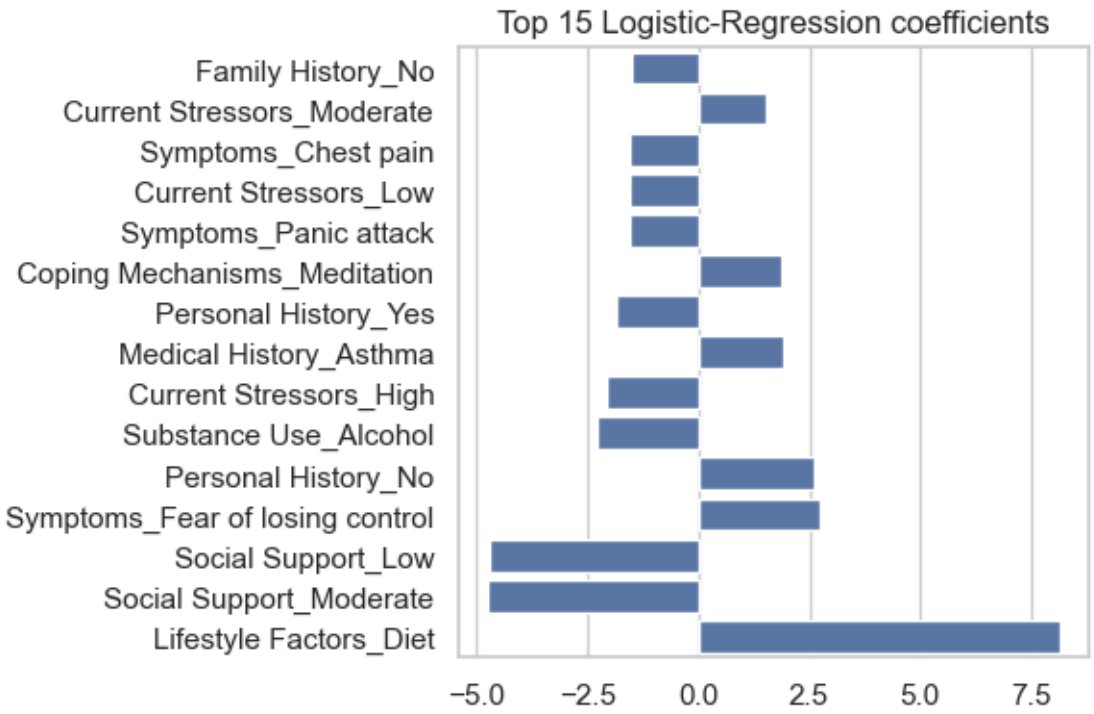
```
coef = log_best.named_steps["clf"].coef_[0]
top_idx = np.argsort(np.abs(coef))[-15:]

plt.figure(figsize=(6, 4))
sns.barplot(x=coef[top_idx], y=feature_names[top_idx])
plt.title("Top 15 Logistic-Regression coefficients")
plt.tight_layout()
plt.show()
```

```
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
```




```
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
```



[46]:
```
high_corr = corr_s.abs().where(~np.eye(len(corr_s), dtype=bool)).stack()
high_corr = high_corr[high_corr > 0.8]
print("\nPotentially redundant numeric pairs (>0.80 abs corr):")
print(high_corr if not high_corr.empty else "None found")
```

```
Potentially redundant numeric pairs (>0.80 abs corr):
None found
```

[55]:
```
from imblearn.over_sampling import RandomOverSampler

def pipe_no_resample(estimator):
    return Pipeline(steps=[("pre", preprocess), ("clf", estimator)])

# LR + SMOTE
lr_smote = make_pipe(LogisticRegression(max_iter=500, solver="liblinear", C=1))
lr_smote.fit(X_train, y_train)
sm_pred = lr_smote.predict(X_test)
sm_auc  = roc_auc_score(y_test, lr_smote.predict_proba(X_test)[:, 1])

# LR + class_weight
lr_cw = pipe_no_resample(
    LogisticRegression(max_iter=500, solver="liblinear",
                       C=1, class_weight="balanced"))
```

```
lr_cw.fit(X_train, y_train)
cw_pred = lr_cw.predict(X_test)
cw_auc  = roc_auc_score(y_test, lr_cw.predict_proba(X_test)[:, 1])

# LR + RandomOverSampler
lr_ros = pipe_no_resample(
    LogisticRegression(max_iter=500, solver="liblinear", C=1))
ros_pipe = Pipeline([("pre", preprocess),
                     ("ros", RandomOverSampler(random_state=42)),
                     ("clf", lr_ros.named_steps["clf"])])
ros_pipe.fit(X_train, y_train)
ros_pred = ros_pipe.predict(X_test)
ros_auc  = roc_auc_score(y_test, ros_pipe.predict_proba(X_test)[:, 1])

# comparison table
comp = pd.DataFrame({
    "Accuracy": [accuracy_score(y_test, p) for p in (sm_pred, cw_pred,␣
 ↪ros_pred)],
    "Recall":   [recall_score(y_test, p)   for p in (sm_pred, cw_pred,␣
 ↪ros_pred)],
    "F1":       [f1_score(y_test, p)       for p in (sm_pred, cw_pred,␣
 ↪ros_pred)],
    "AUC":      [sm_auc, cw_auc, ros_auc]
}, index=["LR + SMOTE", "LR + class_weight", "LR + RandomOS"]).round(3)

display(comp)
```

```
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
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  warnings.warn(
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packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/utils/extmath.py:205: RuntimeWarning: divide by zero
encountered in matmul
  ret = a @ b
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/utils/extmath.py:205: RuntimeWarning: overflow encountered in
matmul
  ret = a @ b
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/utils/extmath.py:205: RuntimeWarning: invalid value encountered
```

```
in matmul
  ret = a @ b
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
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  warnings.warn(
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packages/sklearn/utils/extmath.py:205: RuntimeWarning: divide by zero
```

```
encountered in matmul
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matmul
  ret = a @ b
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/utils/extmath.py:205: RuntimeWarning: invalid value encountered
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  ret = a @ b
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
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packages/sklearn/utils/extmath.py:205: RuntimeWarning: invalid value encountered
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packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
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/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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matmul
  ret = a @ b
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/utils/extmath.py:205: RuntimeWarning: invalid value encountered
in matmul
```

```
  ret = a @ b
```

	Accuracy	Recall	F1	AUC
LR + SMOTE	0.915	0.998	0.502	0.965
LR + class_weight	0.909	1.000	0.486	0.965
LR + RandomOS	0.910	1.000	0.487	0.965

[51]:
```python
from imblearn.over_sampling import RandomOverSampler
from sklearn.metrics import (accuracy_score, recall_score,
                             f1_score, roc_auc_score)
import pandas as pd

# ----------------------------------------------------------------------
# Helpers
# ----------------------------------------------------------------------
def pipe_no_resample(estimator):
    """ Pipeline(preprocess → clf) without SMOTE """
    return Pipeline(steps=[("pre", preprocess), ("clf", estimator)])

def score_model(est, name):
    """ Fit estimator, return dict of test-set metrics """
    est.fit(X_train, y_train)
    y_pred = est.predict(X_test)
    y_prob = est.predict_proba(X_test)[:, 1]
    return {
        "Model": name,
        "Accuracy":  accuracy_score(y_test, y_pred),
        "Recall":    recall_score(y_test, y_pred),
        "F1":        f1_score(y_test, y_pred),
        "AUC":       roc_auc_score(y_test, y_prob)
    }

rows = []
```

[52]:
```python
rf_smote = make_pipe(RandomForestClassifier(random_state=42,
                                            n_estimators=400,
                                            max_depth=None))
rows.append(score_model(rf_smote, "RF + SMOTE"))

rf_cw = pipe_no_resample(
    RandomForestClassifier(random_state=42,
                           n_estimators=400,
                           max_depth=None,
                           class_weight="balanced"))
rows.append(score_model(rf_cw, "RF + class_weight"))

rf_ros = Pipeline([
    ("pre", preprocess),
```

```
    ("ros", RandomOverSampler(random_state=42)),
    ("clf", RandomForestClassifier(random_state=42,
                                   n_estimators=400,
                                   max_depth=None))
])
rows.append(score_model(rf_ros, "RF + RandomOS"))
```

```
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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  warnings.warn(
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observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
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observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
```

```
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
```

[53]:
```
knn_smote = make_pipe(KNeighborsClassifier(n_neighbors=15,
                                           weights="distance"))
rows.append(score_model(knn_smote, "KNN + SMOTE"))

knn_ros = Pipeline([
    ("pre", preprocess),
    ("ros", RandomOverSampler(random_state=42)),
    ("clf", KNeighborsClassifier(n_neighbors=15,
                                 weights="distance"))
])
rows.append(score_model(knn_ros, "KNN + RandomOS"))

knn_none = pipe_no_resample(
    KNeighborsClassifier(n_neighbors=15, weights="distance"))
rows.append(score_model(knn_none, "KNN (no balance)"))
```

```
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
```

```
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
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is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
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is needed for imputation with strategy='median'.
  warnings.warn(
/Users/abirgolam/Downloads/myenv/lib/python3.13/site-
packages/sklearn/impute/_base.py:598: UserWarning: Skipping features without any
observed values: ['Severity' 'Impact on Life']. At least one non-missing value
is needed for imputation with strategy='median'.
  warnings.warn(
```

[54]:
```
df_balance = (pd.DataFrame(rows)
              .set_index("Model")
              .round(3)
              .sort_values("AUC", ascending=False))
display(df_balance)
```

```
                  Accuracy  Recall     F1    AUC
Model
RF + SMOTE           0.970   0.668  0.655  0.987
RF + class_weight    0.969   0.518  0.592  0.987
RF + RandomOS        0.970   0.694  0.665  0.987
KNN (no balance)     0.969   0.354  0.495  0.971
KNN + SMOTE          0.897   0.943  0.439  0.967
KNN + RandomOS       0.918   0.903  0.484  0.961
```